

Target Tracker System Overview

This document summarizes the Java/Swing target tracking application, the scenario generation workflow, the IMM tracker, the track stitching analysis, and the current online spatial birth-density estimator used by the Alternative Hypothesis scoring path.

High-level architecture

Area	Primary responsibility	Representative packages
Scenario model	Maintains targets, paths, velocity profiles, altitude profiles, blackout regions, and preset scenarios.	com.targettracker.model
World view UI	Displays the Plate Carree earth view, grid, truth, measurements, tracks, covariance, replay timeline, and stitching focus modes.	com.targettracker.ui
Sensor and measurements	Applies sensor look timing, offsets, detection probability, Gaussian ECEF measurement errors, and blackout exclusions.	com.targettracker.ui
Tracking	Runs the IMM tracker with CV and CA model options, process noise, association gating, track aging, and covariance limits.	com.targettracker.tracking
Recording and replay	Writes and reads MATLAB-friendly CSV files for ground truth, measurements, tracks, and analysis exports.	com.targettracker.recording
Track stitching analysis	Builds old/new tracklet candidates, estimates join times, scores stitch feasibility, solves assignments, and exports full diagnostics.	com.targettracker.analysis

Main UI view

- Scenario Generation mode is used to draw targets, select preset scenarios, add blackout regions, configure sensor and IMM parameters, pre-compute the scenario, and replay it.
- Analysis Mode loads a previously recorded scenario folder, restores the map region, toggles ground truth and measurements, and enables Track Stitching Analysis.
- Track Stitching Analysis reuses the world view, adds candidate timestamp tabs, shows a red timeline marker for each candidate, and provides All, Greyed, and Only Stitched focus modes.
- The right-side analysis panel shows candidate old/new tracks, truth association, join-time estimates, NLL, Mahalanobis distance, static NLLR, learned-spatial NLLR, and optimal assignments.

Scenario and measurement pipeline

Targets are represented in ECEF state space. User-drawn or preset trajectories are defined through lat/lon paths, altitude profiles, and speed profiles, then converted onto the earth surface geometry so altitude follows the requested height above the ellipsoid-like earth model. During pre-compute, the measurement engine advances through sensor looks and creates Gaussian position and velocity measurements when a target is detectable.

Concept	Current behavior
Measurement timing	A look interval and look offset define scan times, for example 15 s interval and 5 s offset gives 5, 20, 35, ... seconds.
Measurement noise	Position and velocity errors are sampled in ECEF coordinates from zero-mean Gaussian covariance built from the UI standard deviations.
Detection probability	Each target look uses a weighted coin flip from P_d before producing a measurement.
Blackout regions	Rectangular lat/lon regions suppress measurements when a target is inside them. Presets and user-created regions share the same measurement exclusion behavior.
Recording	A recorded run creates a timestamped scenario folder with ground truth, track data, and measurement data subdirectories.

Tracker pipeline

- IMM models currently include CV and CA options. Both are enabled by default and use discrete continuous white-noise acceleration style process-noise construction.
- Measurements are associated to tracks by a greedy solve under a Mahalanobis threshold.
- Between updates, the display coasts tracks by prediction from the last update state and time. The stored state is not repeatedly overwritten while coasting.
- Tracks can be declared dead by timeout or covariance radius threshold. Dead tracks remain visible as grey tracks in analysis views.
- Saved track rows contain track id, time, 9D state, 9x9 covariance, and an updated/coasted Boolean at each second mark.

Track stitching analysis

The stitching analysis searches measurement times where old tracks are coasting and new tracks have recently formed. It compares each old tracklet i with each new tracklet j , estimates possible join times, scores the state innovation at those times, and optionally solves one-to-one assignments using the Hungarian algorithm.

Output	Meaning
Simple midpoint	Halfway between the old anchor time and new anchor time.
Kinematic midpoint	Uses old/new spatial separation and average speed to place a travel-time-informed midpoint.
Statistical bank time	Predicts old and retrodicts new over a bank of times, computes the innovation and innovation covariance, then selects the midpoint of the two best Mahalanobis times.
Truth time	Diagnostic-only truth reference. It is not allowed in optimal assignment because it is unavailable in real data.
NLL	Canonical Gaussian negative log likelihood of the old/new innovation.
Mahalanobis	Innovation transpose times inverse innovation covariance times innovation.
Static NLLR	NLL compared against a uniform false-alarm plus birth density over the innovation volume.
Learned-spatial NLLR	NLL compared against the online learned birth-density estimate over the innovation volume.

Prediction and retrodiction notes

- Old tracklets are predicted forward from their last good update state, even when the old track is already dead.
- New tracklets are retrodicted backward from a future anchor using negative kinematic dt but positive covariance growth.
- Earlier measurements on the new tracklet are assimilated while retrodicting, producing a smoothing-like backward estimate before extrapolating toward the join time.

Online spatial birth-density estimator

The learned Alternative Hypothesis path now uses an online evidence/exposure grid. The unit is births per scan per cubic kilometer. This is intentionally matched to the measurement-time update loop: every scan adds exposure, and matured first-measurement track starts add positive birth evidence.

Step	Implementation
Spatial cells	The operating ECEF bounds are padded and divided into bounded 3D cells. Cell size is derived from scenario span, clamped between 0.5 km and 5 km.
Weak prior	Every cell starts with prior exposure and prior evidence from the configured birth rate. This avoids unstable zero-density or infinite-ratio behavior.
Exposure	At each measurement scan, observable cells receive exposure in scan-km ³ units. With the current God sensor assumption, the full operating grid is observable.
Candidate births	Only the first associated measurement for each track can seed birth evidence. Later updates from that track never add new birth evidence.
Ambiguity delay	Birth candidates are held until the configured new-track window has elapsed. This gives the stitcher time to explain the start before it becomes evidence.
Confidence	A birth near a plausible predecessor is downweighted; isolated starts receive higher confidence.
Deposit	Birth evidence is deposited over nearby cells with a Gaussian weight based on cell size and measurement uncertainty.
Forgetting	Both evidence and exposure decay with a 30 minute half-life so old traffic patterns fade.
Density	Cell density equals birth evidence divided by observation exposure, clamped to configured min and max values.
Query	A stitching query smooths nearby cells over the innovation-volume radius and state uncertainty, then returns expected births, local density, exposure, and reliability.

Query math

For a track-to-track innovation at candidate join time t , the analysis computes an innovation covariance S . The position block of S defines an ellipsoid-like volume V in km³. The density estimator returns ρ_{hat} near the midpoint of the old predicted and new retrodicted states. The learned extraneous count is $\lambda_{\text{ex}} = \max(\epsilon, \rho_{\text{hat}} * V)$. The learned-spatial NLLR reported in the UI and export is:

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learned_NLLR = NLL + log( lambda_ex )
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Reliability is derived from exposure. Low reliability means the answer is mostly prior-driven; high reliability means repeated observable scans have shaped the local density.

MATLAB-friendly outputs

Folder or file	Contents
ground truth data	Target id, time, and ECEF state for each target.
track data	Track id, time, 9D state, 9x9 covariance, and updated/coasted Boolean at each second.
measurement data	Sensor id, time, mean, covariance, and measurement uncertainty.
analysis summary	Events, segments, pair time estimates, and optimal assignments.
analysis bank_evaluations	Every old/new state pair over the time bank, including innovation, covariance, NLL, NLLR, density query, exposure, and reliability.
analysis spatial_density	Scenario-time history of density grid summary values: representative density, peak density, mean density, evidence, exposure, reliability, cell size, and grid shape.

Practical interpretation

- Airport and hangar scenarios should accumulate high local density because many unrelated first measurements occur in nearby cells.
- A one-target hard turn should not look like a busy birth region because it contributes at most one matured birth and many scans of exposure.
- If a region has low exposure, the estimator intentionally stays closer to the prior and reports lower reliability.
- If a new track is later stitchable to an old one, the confidence gate and maturation delay reduce the chance that it pollutes the birth field.